Making Every Hour Count: Turning Sleep Data into Better Rest

Team 6
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Introduction

Introduction

Motivation

- Limited sleep time makes sleep efficiency important.
- Sleep efficiency = Time asleep / Time in bed.

Research Goal

- Estimate sleep efficiency without specialized sleep-monitoring devices.
- Use self-reportable lifestyle and sleep-related features.
- Investigate interpretable regression models.



Data Distributions & Descriptive Statistics



Data Distributions & Descriptive Statistics

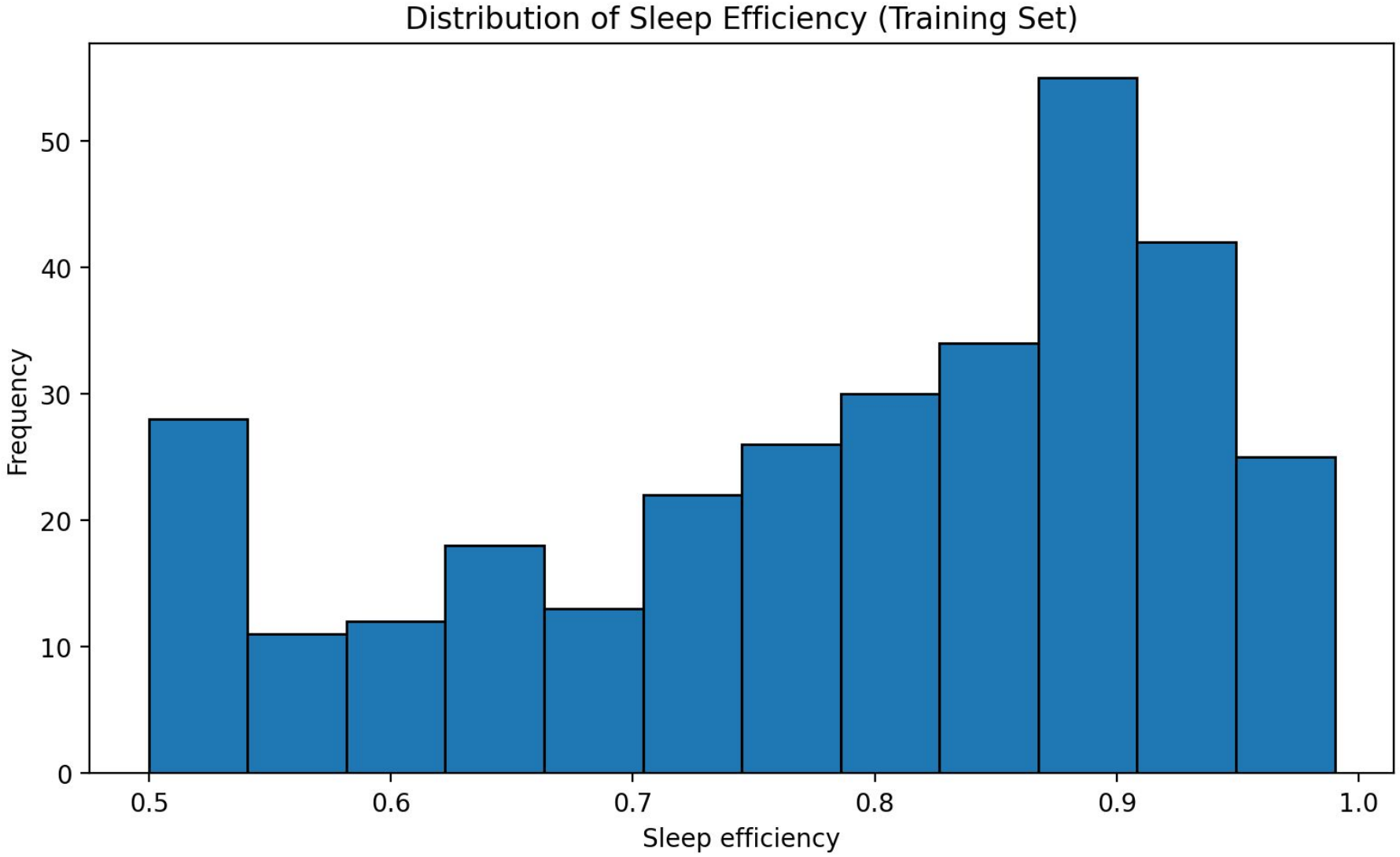
Descriptive Statistics

- Mean sleep efficiency: 0.79 ± 0.14 (SD).
- Slightly left-skewed distribution.
- Some missing values in lifestyle and sleep-related features.

Data Preparation

- 70/30 train-test split.
- EDA performed on training data only to avoid data leakage.

Data Distributions & Descriptive Statistics



Bivariate

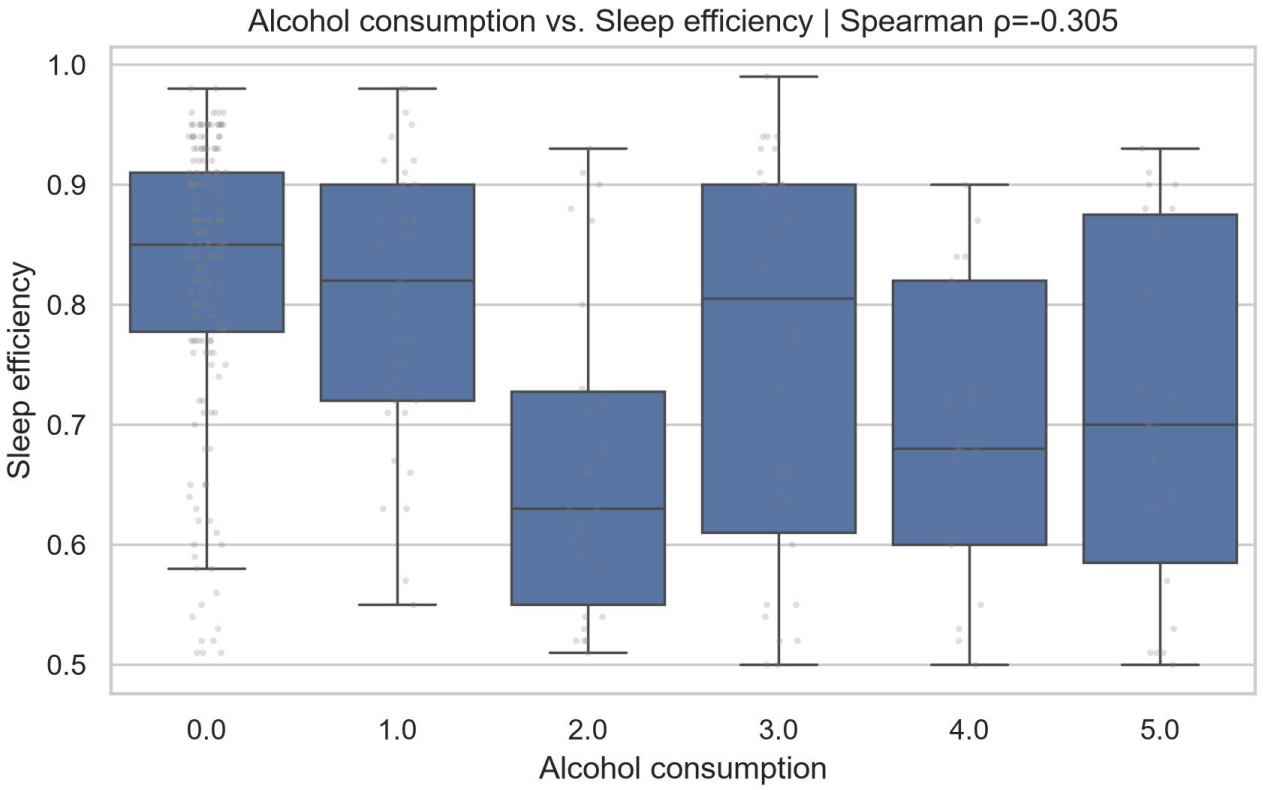


Feature v.s. Sleep efficiency

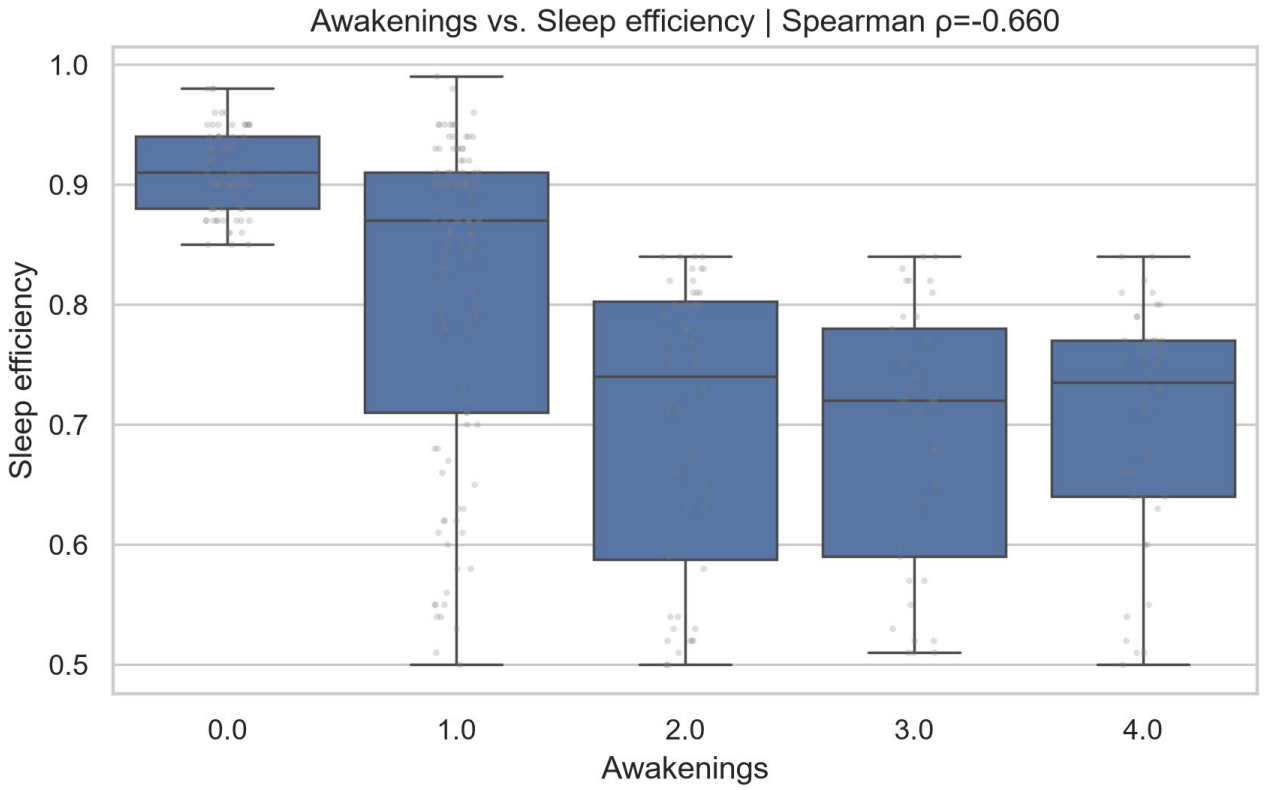
Negative impacts are more significant than positive impacts

Predictor	Pearson r	p	Spearman ρ	p
Awakenings	-0.550	< 0.001	-0.660	< 0.001
Alcohol consumption	-0.328	< 0.001	-0.305	< 0.001
Exercise frequency	0.243	< 0.001	0.255	< 0.001
Age	0.158	0.0048	0.154	0.0062
Caffeine consumption	0.091	0.1151	0.056	0.3352
Sleep duration	-0.034	0.5515	-0.025	0.6546

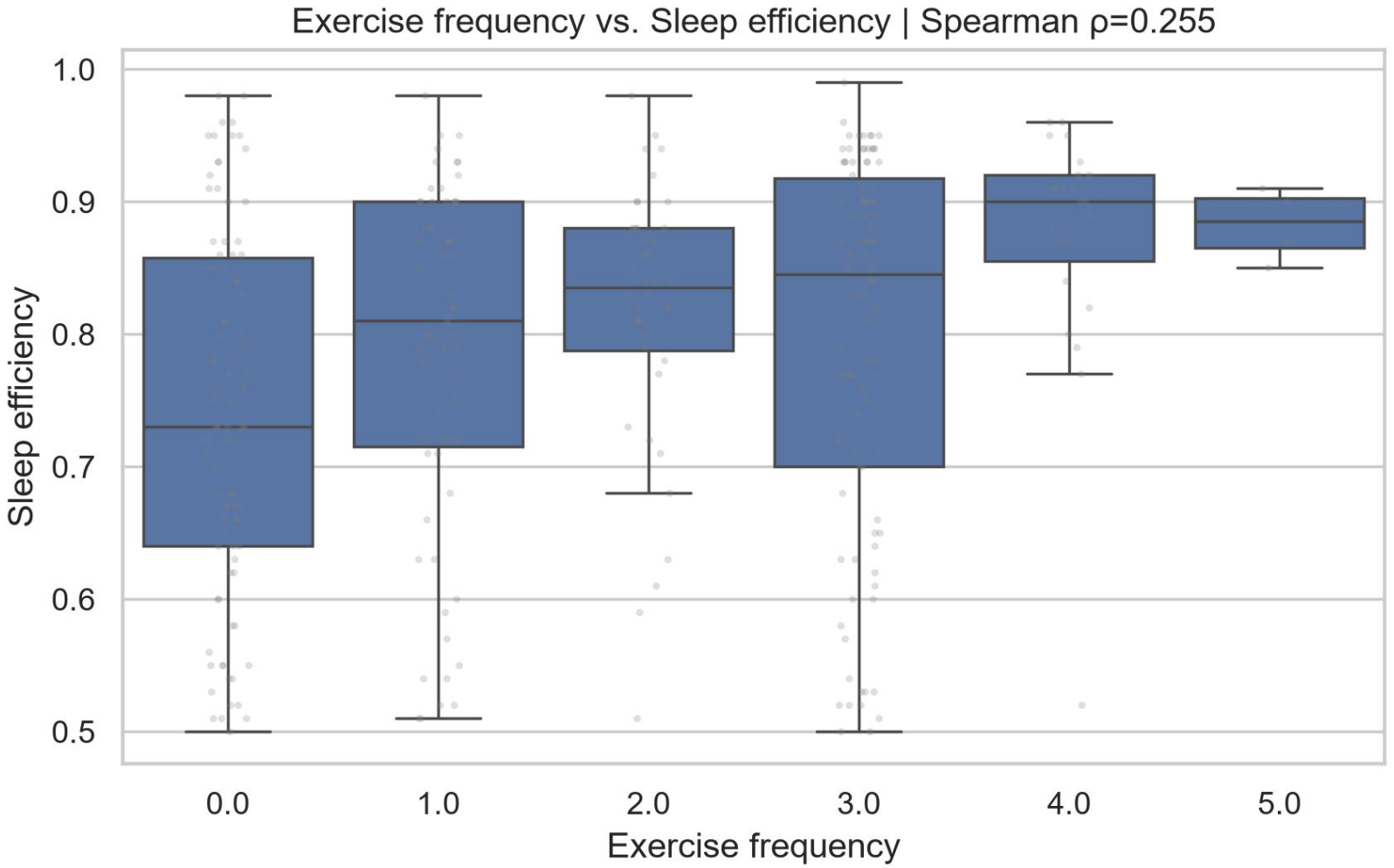
Alcohol : Pearson's R = -0.550



Awakenings : Pearson's R = -0.328



Exercise frequency : Pearson's R = 0.243

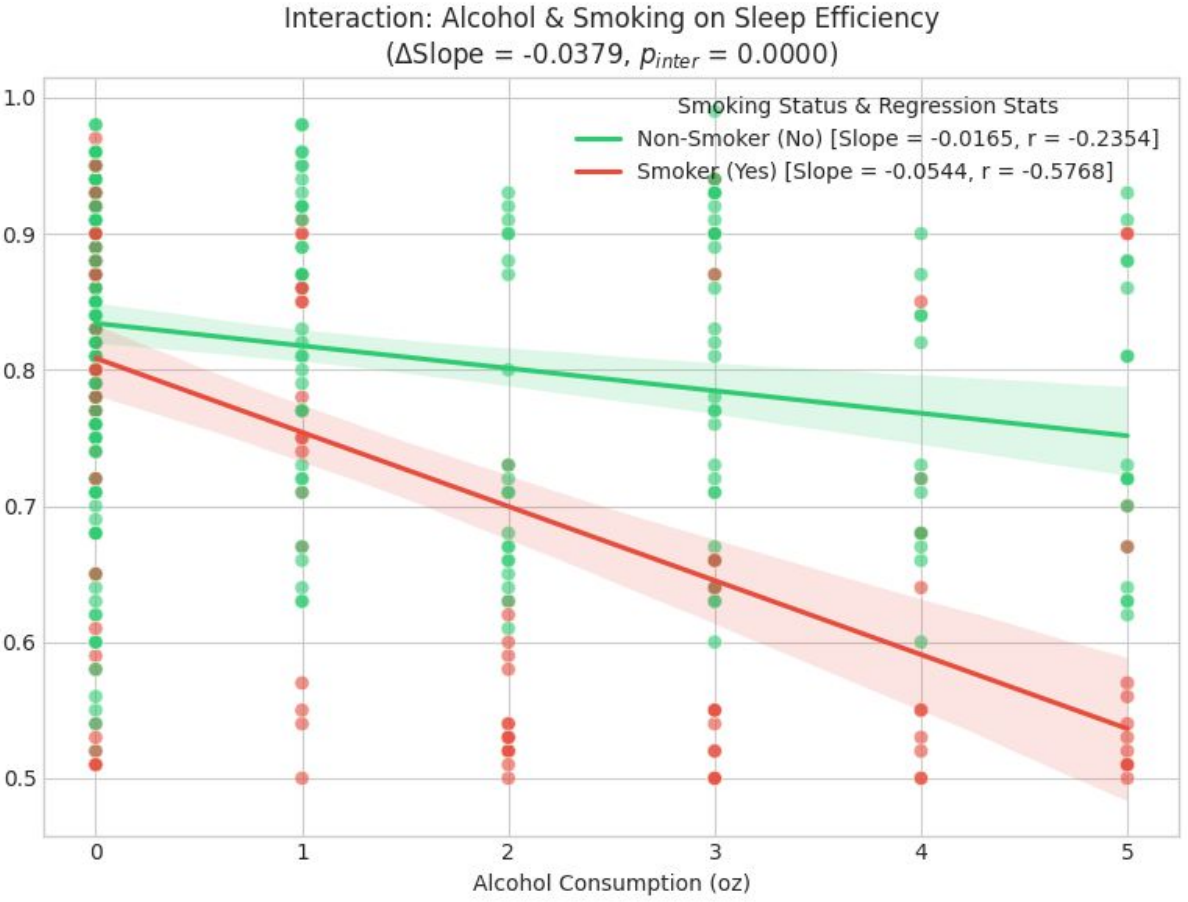


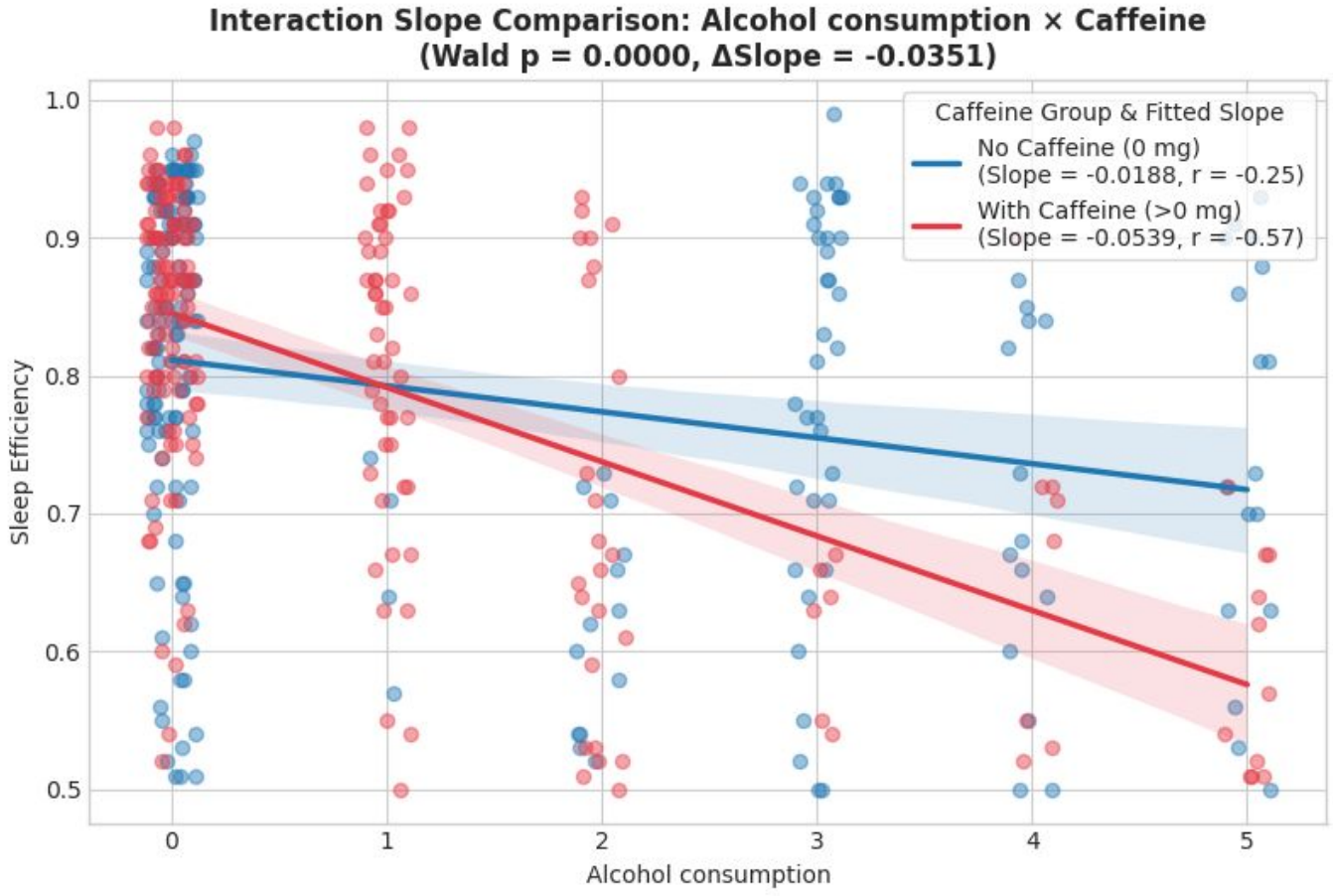
Feature Interaction



Feature Interactions

Most of them are not significant (either *slope_p > 0.05* or no different between groups)





Alcohol x Smoking

Alcohol x Caffeine

Sleep duration x Bedtime

Age x Bedtime

Sleep duration x Caffeine

Awakening x Caffeine

Exercise frequency x Caffeine

Test	Group	Slope	CI_low	CI_high	Slope_p
Alcohol x smoking status	Yes	-0.04831581910302100	-0.0656192543273023	-0.031012383878739700	4.43120880596562E-08
Alcohol x smoking status	No	-0.01115963270694200	-0.02065036752136980	-0.0016688978925141000	0.021188213814092500
Alcohol x caffeine (binary >0 mg)	0.0	-0.011630823305135200	-0.024654361047307700	0.0013927144370373000	0.08005468738334370
Alcohol x caffeine (binary >0 mg)	1.0	-0.050027860242064400	-0.062092697227233200	-0.03796302325689560	4.39482035953025E-16
Alcohol x caffeine (3 levels)	None	-0.011630823305134600	-0.024654361047307100	0.0013927144370377900	0.08005468738335670
Alcohol x caffeine (3 levels)	Low (0-50 mg]	-0.04661446533756390	-0.05913451539644090	-0.034094415278686900	2.93580178262859E-13
Alcohol x caffeine (3 levels)	High (>50 mg)	-0.14322580645161200	-0.21279369389177900	-0.07365791901144560	5.45655504512691E-05
Sleep duration x bedtime (4 groups)	Late (00:00-05:59)	-0.0014286573266788600	-0.025319195491614300	0.02246188083825660	0.9066967297772160
Sleep duration x bedtime (4 groups)	Early (18:00-22:59)	0.009493510604621100	-0.028203394892611500	0.04719041610185370	0.6215934543407920
Sleep duration x bedtime (4 groups)	Regular (23:00-23:59)	-0.03704761904761930	-0.08194514728372950	0.007849909188490910	0.10581725574841900
Age x caffeine (binary)	0.0	0.0033274996045159200	0.0017676951901748500	0.004887304018856980	2.90035858741116E-05
Age x caffeine (binary)	1.0	0.0001160186824202200	-0.0015787251834796 0	0.001810909883188400	0.8932733327398740
Age x caffeine (3 levels)	None	0.003327499604516190	0.001767695190175090	0.004887304018857290	2.90035858740798E-05
Age x caffeine (3 levels)	Low (0-50 mg]	8.44713992756928E-05	-0.0017212847528874100	0.0018902275514387900	0.9269482505580940
Age x caffeine (3 levels)	High (>50 mg)	0.0008669331139044380	-0.0029721942367803100	0.004706060464589190	0.658062638378003
Sleep duration x caffeine (binary)	0.0	-0.01981910411830430	-0.04728114846037380	0.007642940223765090	0.15721868858319300
Sleep duration x caffeine (binary)	1.0	0.00673656351791526	-0.018977740803850300	0.03245086783968080	0.6076253639666220
Sleep duration x caffeine (3 levels)	None	-0.01981910411830450	-0.04728114846037400	0.007642940223764900	0.1572186885831890
Sleep duration x caffeine (3 levels)	Low (0-50 mg]	0.010692591730148400	-0.0170485050093225	0.038433688469619200	0.4499772241416640
Sleep duration x caffeine (3 levels)	High (>50 mg)	-0.004316546762588730	-0.05058362631985600	0.04195053279467860	0.8549099376273070
Awakenings x caffeine (binary)	0.0	-0.05268844288432180	-0.06575894010793440	-0.03961794566070930	2.77107775489305E-15
Awakenings x caffeine (binary)	1.0	-0.06165544462502920	-0.07304848862555490	-0.05026240062450350	2.77411897491023E-26
Awakenings x caffeine (3 levels)	None	-0.052688442884321400	-0.06575894010793390	-0.039617945660708900	2.77107775489399E-15
Awakenings x caffeine (3 levels)	Low (0-50 mg]	-0.06281235097758690	-0.0753192904860222	-0.050305411469151700	7.32447566612776E-23
Awakenings x caffeine (3 levels)	High (>50 mg)	-0.04690201729106600	-0.07466708223249650	-0.019136952349635500	0.0009300946833980900
Exercise frequency x caffeine (binary)	0.0	0.026292908153358400	0.012020056424113500	0.04056575988260330	0.00030552294958342700
Exercise frequency x caffeine (binary)	1.0	0.029907494145199400	0.015391239193623400	0.044423749096775400	5.38934411877614E-05
Exercise frequency x caffeine (3 levels)	None	0.026292908153358000	0.012020056424113 20	0.04056575988260290	0.0003055229495834780
Exercise frequency x caffeine (3 levels)	Low (0-50 mg]	0.028139590092584800	0.013115587901520000	0.04316359228364960	0.0002416366250209 6100
Exercise frequency x caffeine (3 levels)	High (>50 mg)	0.0022289156626503100	-0.06675947254656040	0.07121730387186110	0.9495088081113550

Method: Pipeline and Model

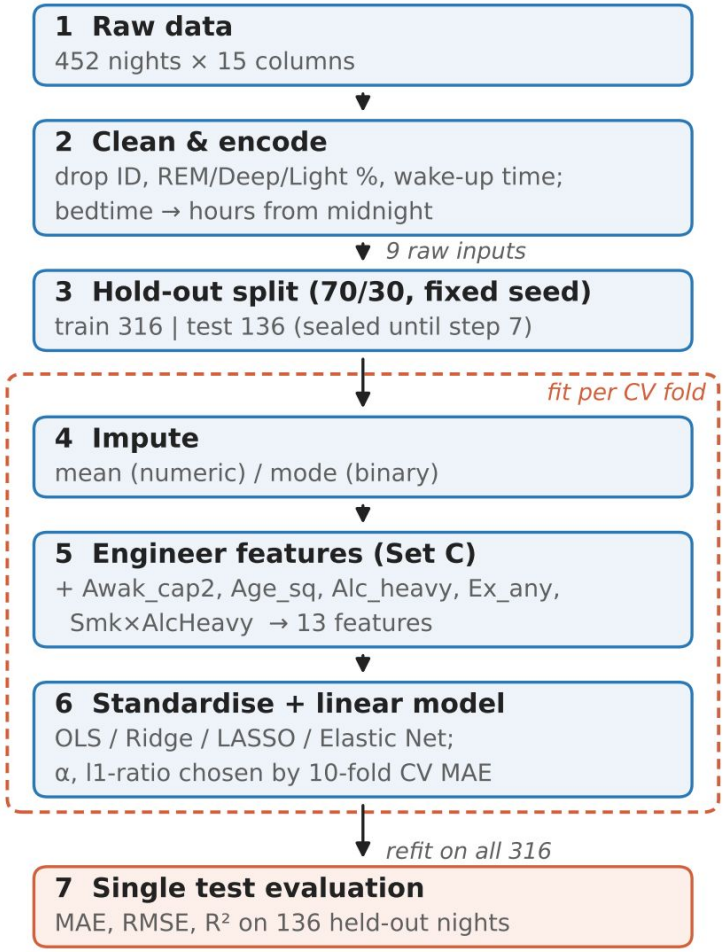


Figure 3. Modelling pipeline

- Nine raw inputs after exclusions.
- Fit imputation, feature construction, and scaling inside each CV training fold.
- Compare OLS, Ridge, LASSO, and Elastic Net.
- Retain Elastic Net: $\alpha = 0.0017$, L1 ratio = 0.9.

Method: Engineered Features

Feature	Definition	EDA rationale
Awak_cap2	min(awakenings, 2)	Efficiency plateaus after two awakenings
Age_sq	$(\text{age} - 40)^2$	Concave age profile
Alc_heavy	1 if alcohol ≥ 2	A clear drop from two drinks
Ex_any	1 if exercise ≥ 1	Main gain: none versus any exercise
Smk × AlcH	smoker × Alc_heavy	Lower efficiency with joint exposure

- Awak_cap2 replaces the raw count. The other four features add columns.
- Set A: 9 raw inputs. Set C: 13 features.

Challenges

- **Non-linear Plateau in Awakenings**
  - Raw awakening count biased linear models at high values as sleep efficiency plateaus after 2 awakenings
  - Capped feature via Awak_cap2 = min(awakenings, 2).
- **Concave Age Relationship**
  - Standard linear terms failed to reflect non-monotonic age dynamics centered around age 40 Engineered
  - quadratic feature Age_sq = $(age - 40)^2$.
- **Threshold & Interaction Non-linearities**
  - Alcohol impact exhibits a abrupt step-drop at 2 drinks with compound negative effects when combined with smoking
  - Added binary threshold Alc_heavy and interaction Smk AlcH.

Evaluation: Protocol and Baselines

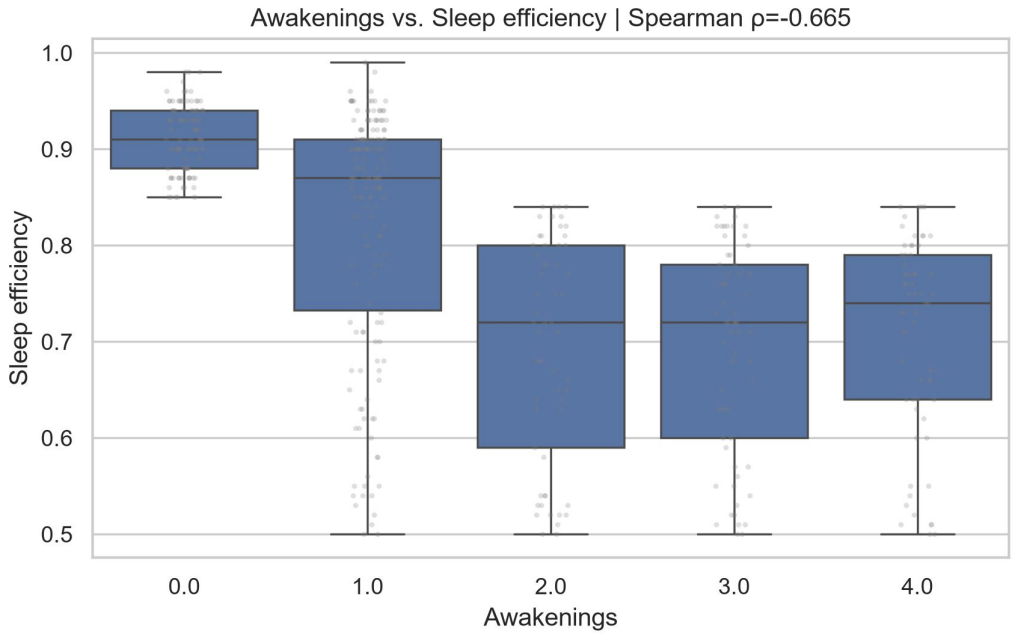
- 70/30 random split: 316 training and 136 test nights. Seed: 20261010.
- Shuffled 10-fold CV with identical folds for every method.
- Tune on training CV, refit on all training rows, then evaluate test once.
- Metrics: MAE (primary), RMSE, and R^2. A 0.01 error is one percentage point.
- Decision target: efficiency 0.85. Tolerance: MAE $\leq$ 0.05.
- Baselines: training mean, awakenings-only OLS, Random Forest, and Gradient Boosting.

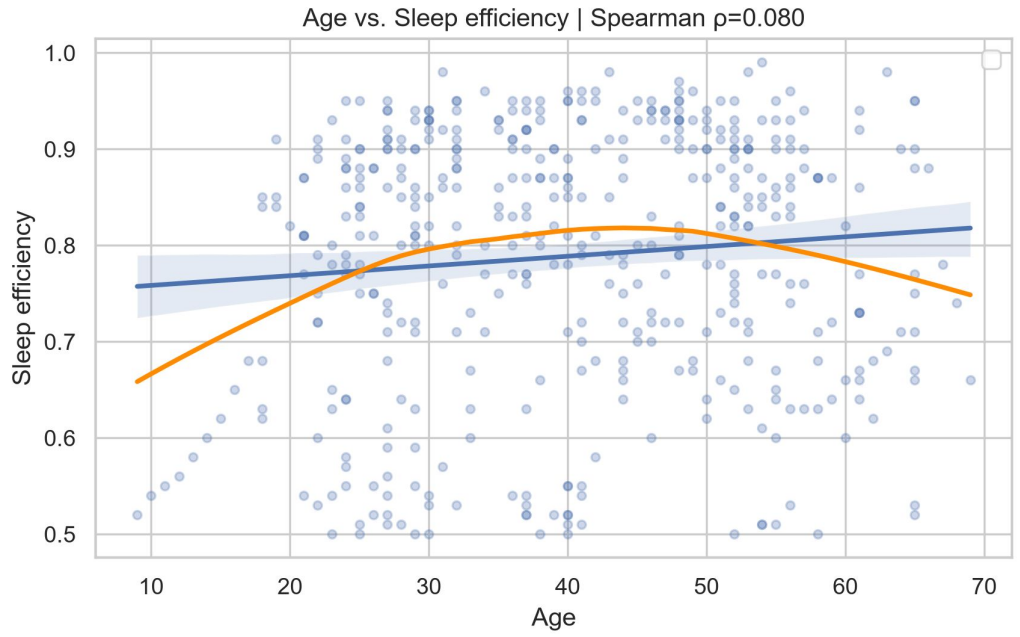
Evaluation: Main Results

Model	CV MAE	Test MAE	Test RMSE	Test R^2
Training mean	0.115 ± 0.015	0.115	0.134	−0.00
Awakenings-only OLS	0.091 ± 0.012	0.090	0.108	0.34
Set A: OLS (raw)	0.081 ± 0.009	0.074	0.093	0.52
Set B: OLS (top-6)	0.080 ± 0.009	0.074	0.093	0.51
Set C: Elastic Net	**0.069 ± 0.009**	**0.063**	**0.081**	**0.63**
Random Forest	**0.061 ± 0.006**	**0.059**	**0.078**	**0.66**
Gradient Boosting	0.064 ± 0.009	0.061	0.079	0.65

- Set C lowers test MAE by 15% versus raw OLS. Random Forest performs best.
- Set C misses the 0.05 tolerance. 51% of test nights fall within ±0.05.

Evaluation: Awakenings and Age





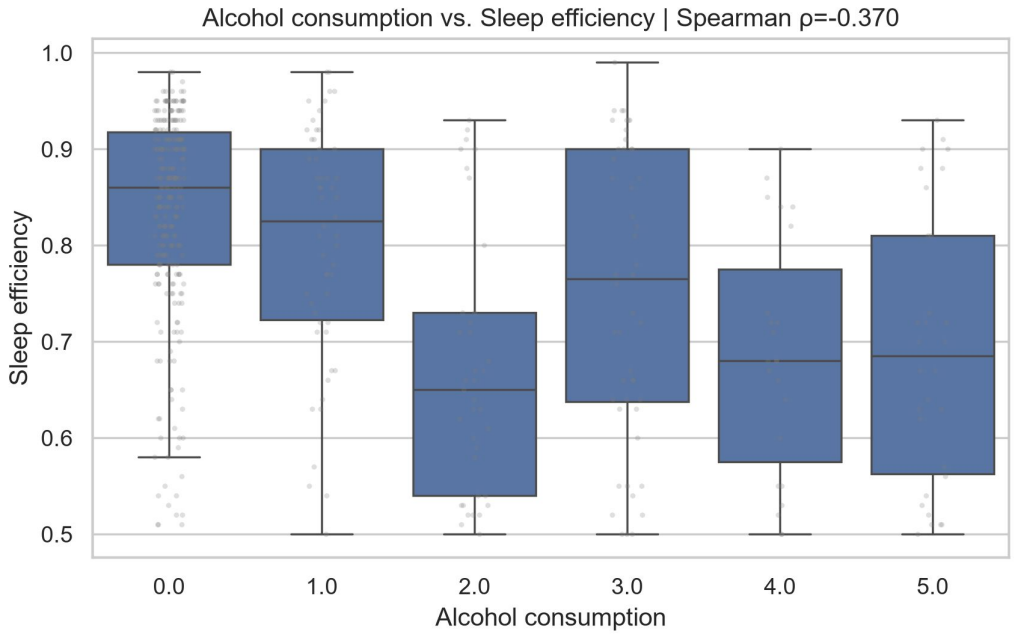
Awak_cap2 = min(awakenings, 2)

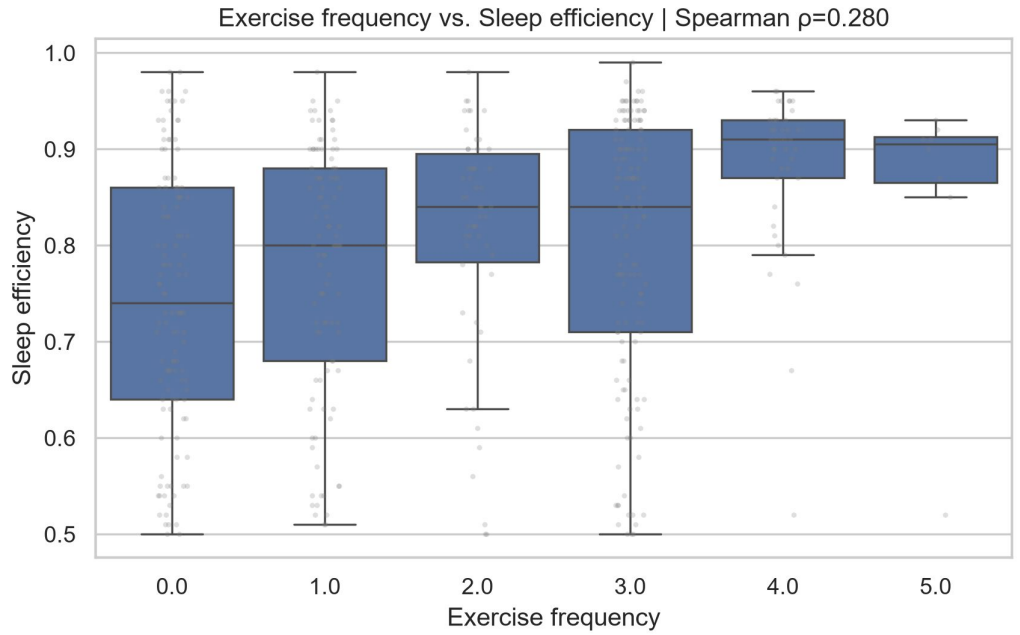
- Mean efficiency: 0.91 (0), 0.81 (1), about 0.70 (≥2).
- The cap represents the plateau after two awakenings.

Age_sq = (age − 40)²

- Weak linear correlation, but a concave age profile.
- Quadratic relationship: HC3 robust p < 0.001.

Evaluation: Alcohol and Exercise





Alc_heavy = 1 if alcohol ≥ 2

- One drink differs little from none.
- The report uses ≥2 to encode the observed drop.

Ex_any = 1 if exercise ≥ 1

- The report emphasizes none versus any exercise.
- The indicator captures a step beyond the raw count.

Evaluation: Feature Ablation

Feature set (OLS)	Features	CV MAE	Δ vs. A
A: raw inputs	9	0.0808 ± 0.009	—
B: top-6 by |r|	6	0.0797 ± 0.009	−1.4%
A, capped awakenings	9	0.0780 ± 0.008	−3.4%
+ Age_sq	10	0.0772 ± 0.007	−4.4%
+ Alc_heavy	11	0.0755 ± 0.008	−6.5%
+ Ex_any	12	0.0743 ± 0.008	−8.0%
+ Smk × AlcH = C	**13**	**0.0692 ± 0.010**	**−14.3%**

- Smokers drinking ≥2: mean efficiency 0.60, versus 0.79–0.83 in the other groups.
- Smk × AlcH captures joint exposure and gives the largest cumulative MAE drop.

Interpretation: Key Coefficients

OLS on standardized Set C (n = 316). β: efficiency change per one SD.

Feature	β	95% CI	p-value	VIF
Awak_cap2	−0.066	[−0.076, −0.056]	<0.001	1.1
Smk × AlcH	−0.042	[−0.057, −0.028]	<0.001	2.3
Alc_heavy	−0.027	[−0.051, −0.003]	0.030	6.4
Ex_any	+0.023	[+0.006, +0.040]	0.007	3.0
Age	+0.019	[+0.009, +0.030]	<0.001	1.2
Age_sq	−0.017	[−0.027, −0.006]	0.002	1.2

- Awakenings and smoking × heavy drinking have the largest negative coefficients.
- Elastic Net shows similar effects and shrinks raw alcohol to zero.

Interpretation: Failure Cases

Actual	Predicted	Error	Input context	Light sleep
0.50	0.80	−0.30	Female, 29, smoker 1 awakening, 1 drink	50%
0.51	0.75	−0.24	Male, 55, smoker 4 awakenings, alcohol missing	53%
0.60	0.79	−0.19	Female, 55, nonsmoker 3 awakenings, 0 drinks	47%

Error = actual − predicted. Light sleep is excluded from model inputs.

- All three largest errors overestimate efficiency.
- Mean alcohol imputation (1.2) keeps case 2 below the heavy-drinking threshold.
- Proposed fix: an alcohol missingness indicator or model-based imputation.

Related Works



Related Works

- **Using wearable sensor data to predict sleep quality**
  Sleep Quality Prediction From Wearable Data Using Deep Learning (Sathyanarayana et al., 2016)
- **Predict low sleep quality hours before bedtime**
  Towards proactively improving sleep: machine learning and wearable device data forecast sleep efficiency 4–8 hours before sleep onset (Sakal et al., 2025)
- **Combining activity, light exposure, and HRV to improve sleep prediction performance**
  Predicting sleep based on physical activity, light exposure, and Heart rate variability data using wearable devices (Park et al., 2024)
- **All works above are classification studies.**

Conclusion

- **Model Performance**
  - 13-feature linear model achieves **MAE = 0.063** and **R^2 = 0.63** without sleep-stage measurements
  - Outperforms raw linear model by 15% and performs comparably to Random Forest
  - Gains are driven by threshold encoding and key feature interaction, rather than complex model choice
- **Key Drivers & Actionable Insights**
  - **Primary Drivers:** Nightly awakenings and the combination of smoking with heavy drinking (2+ drinks)
  - **User Recommendation:** Avoiding 2+ drinks matters most for smokers, who face a **0.13 sleep efficiency penalty**
  - Sleep duration, caffeine, sex, and bedtime contribute surprisingly little
- **Deployment Application**
  - Well-suited for **flagging likely poor sleep nights (97% recall)** rather than precise point estimation.

Contributions

- 林程亮 (112550125):EDA(Alcohol, Gender), Slide & Wrote § 2.
- 林庠安 (315551119):EDA(Smoking), Feature engineering, Wrote § 5 to 9.
- 林景玄 (315551127):EDA(Awakenings), Feature engineering, Slide § 5 to 9.
- 郭佳翰 (315551137):Baselines, Feature Engineering.
- 林李奕 (315553025):EDA(Age, Exercise Frequency), Feature engineering.
- 王裕昕 (315554006):EDA(Overall), Baselines, Feature engineering, Final model.
- 顏丞硯 (315554021):EDA(Bedtime, Wakeup time, Sleep duration), Slide & Wrote § 4.
- 林群佑 (315554038):EDA(Caffeine), Slide & Wrote § 1, 3 and 4, Editorial and analytical revision of the report.



AI usage

- GPT-6.1 Luna: GPT-6.1 Luna used for EDA plot assisting and training baselines.
- Opus 5.5 : Draft and debug analysis code (EDA, the modelling notebook, significance tests and plots), review the pipeline for data leakage, and draft and polish the text of Sections 5–9.
- GPT-6: Used to find related works.
- Gemini 3.6 Flash : mainly use for making plots & Revising sentences
- Gemini 4 Argon: report review
- DeepSeek V4.1 Flash: report review

